

Natural Logarithmic function

(1)

Recall: The log. function is the inverse of the exponential function.

ie: $y = e^x \Rightarrow \boxed{e^y = x}$ inverse

(OR) $\boxed{y = \log_e x}$ (base "e")

log with base "e" is given a "special" name: $y = \log_e x = \underline{\ln(x)}$

↑
read "lawn of x"

Given $y = \ln(x)$ find y' .

* Rewrite $y = \ln(x)$ ^{log form} to be $e^y = x$ ^{exp form.}

* Let $y = g(x)$: $e^{g(x)} = x$

* Differentiate: $e^{g(x)} \cdot g'(x) = 1$

$$g'(x) = \frac{1}{e^{g(x)}} = \frac{1}{e^y} = \frac{1}{x}$$

* $\therefore y' = \frac{1}{x}, x > 0$

In general, if $f(x) = \ln(g(x))$, then

$$f'(x) = \frac{g'(x)}{g(x)}$$

$$\frac{1}{g(x)} \cdot g'(x)$$

why?

$$\log_e e = 1$$

notes:

$$\textcircled{1} \ln e = 1 \rightarrow e' = e *$$

$$\textcircled{2} \ln 1 = 0 \rightarrow e^0 = 1 *$$

$$\ln(x) \rightarrow \frac{1}{x} \cdot x' = \frac{1}{x}$$

$$\log_e 1 = ? \quad e^? = 1$$

$$? = 0$$

Example 1: Differentiate.

$n \neq 0$

$$\ln(n \cdot x) = \frac{1}{x}$$

a) $y = \ln(4x)$

$$y' = \frac{4}{4x}$$

$$y' = \frac{1}{x}$$

b) $y = \ln(x^2 + 5x - 7)$ ← log

$$y' = \frac{2x + 5}{x^2 + 5x - 7}$$
 ← rational.

c) $y = [\ln(x)]^5$

$$y' = 5 [\ln(x)]^4 \cdot \frac{1}{x}$$

$$y' = \frac{5 [\ln(x)]^4}{x}$$

$$\ln(x^5) = 5 \ln(x)$$

$$\frac{5}{x}$$

d) $y = 2x \cdot \ln(x)$

$$y' = (2x)' \ln(x) + 2x \cdot \frac{1}{x}$$

product rule.

$$y' = 2 \ln(x) + 2$$

$$y' = 2 (\ln(x) + 1)$$

crit #s: $\ln(x) + 1 = 0$
 $\ln(x) = -1$
 $x = e^{-1}$
 $x = 1/e$

e)

$$y = \ln \left[\sqrt{\frac{x-1}{x+1}} \right]$$

$$\frac{\left(\sqrt{\frac{x-1}{x+1}} \right)'}{\sqrt{\frac{x-1}{x+1}}}$$

$$y = \ln \sqrt{x-1} - \ln \sqrt{x+1}$$

$$y = \ln (x-1)^{\frac{1}{2}} - \ln (x+1)^{\frac{1}{2}}$$

$$\log \left(\frac{A}{B} \right)$$

$$= \log A - \log B$$

$$y = \frac{1}{2} \ln (x-1) - \frac{1}{2} \ln (x+1)$$

$$y' = \frac{1}{2(x-1)} - \frac{1}{2(x+1)}$$

$$y' = \frac{x+1 - (x-1)}{2(x^2-1)}$$

diff of squares

$$y' = \frac{2}{2(x^2-1)}$$

$$y' = \frac{1}{x^2-1}$$

$$f) y = \ln(\sin(x))$$

5

$$y' = \frac{\cos(x)}{\sin(x)}$$

$$y' = \cot(x)$$

Ex 2: Determine the equation of the normal line to $y = \frac{\ln(x^2)}{3x}$ at $x=1$.

$$y' = \frac{\left(\frac{2x}{x^2}\right)(3x) - \ln(x^2) \cdot 3}{(3x)^2} \quad \frac{2}{3} \left[\frac{\ln x}{x} \right]$$

$$y' = \frac{6 - 3\ln(x^2)}{9x^2} \quad \begin{array}{l} 6\ln x \text{ or } \ln x^6 \\ 6 - 3\ln(1) \\ 9 \end{array}$$

$$\frac{0}{3} = 0 \quad m_t|_{x=1} = \frac{6-0}{9} = \frac{2}{3} \Rightarrow m_{\perp} = -\frac{3}{2}$$

$$\frac{\ln(x^2)}{3x} \quad y - y_1 = m_{\perp}(x - x_1)$$

$$y - 0 = -\frac{3}{2}(x - 1)$$

$$y = -\frac{3}{2}(x - 1)$$